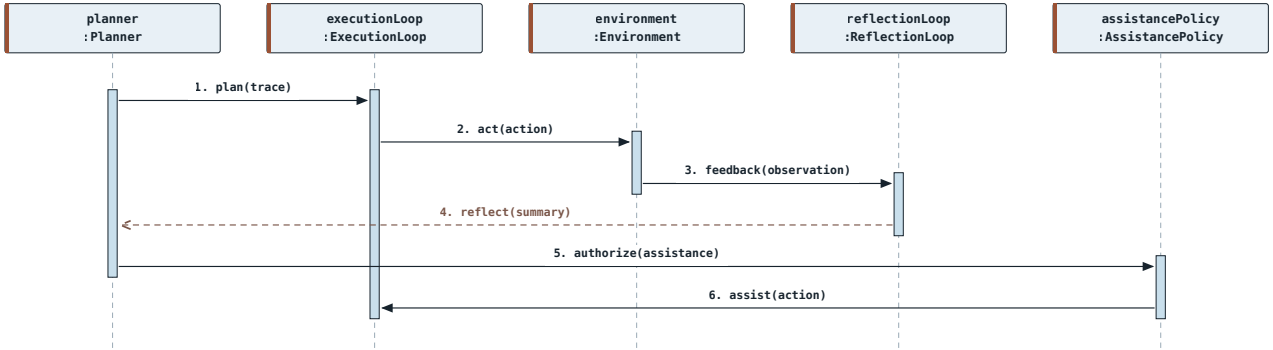
E09 Dual-loop assistance

Evolution / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Separate planning and execution feedback.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	A planner or assistance loop and an executor loop operate concurrently or iteratively.
TRIGGER	Executor state or environment feedback updates planning.
STRUCTURE	Planner, executor, feedback channel, synchronization, and authorization boundary.
EFFECT	The next plan or assistance action changes in response to attributable executor feedback.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Planner and executor states have compatible identities and timing.
ENSURES	Feedback reaches planning before the affected action when required.
INVARIANT	The outer loop cannot bypass executor authorization for physical or external actions.
VIOLATION	Stale, conflicting, or unavailable feedback has a safe coordination policy.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).

Analytical collaboration only; no repository execution or message order is asserted.